



2014 Half-Yearly Examination

# FORM VI

## MATHEMATICS EXTENSION 2

Wednesday 26th February 2014

### General Instructions

- Writing time — 2 hours
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

### Total — 70 Marks

- All questions may be attempted.

### Section I – 10 Marks

- Questions 1–10 are of equal value.
- Record your solutions to the multiple choice on the sheet provided.

### Section II – 60 Marks

- Questions 11–14 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

### Collection

- Write your candidate number on each booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place your multiple choice answer sheet inside the answer booklet for Question Eleven.
- Write your candidate number on this question paper and submit it with your answers.

### Checklist

- SGS booklets — 4 per boy
- Multiple choice answer sheet
- Candidature — 81 boys

Examiner  
RCF

**SECTION I - Multiple Choice**

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

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**QUESTION ONE**

The solutions to the quadratic equation  $(z - 2i)(z - 1 - i) = 0$  are:

**1**

- (A)  $z = 2i$  or  $-1 - i$  (B)  $z = 1 + i$  or  $2i$   
(C)  $z = -2i$  or  $-1 - i$  (D)  $z = 1 + i$  or  $-2i$

**QUESTION TWO**

A primitive of  $e^{2x} \sin(e^{2x})$  is:

**1**

- (A)  $2 \cos(e^{2x})$  (B)  $\frac{1}{2} \sin(e^{2x})$   
(C)  $-2 \sin(e^{2x})$  (D)  $-\frac{1}{2} \cos(e^{2x})$

**QUESTION THREE**

Which of the following represents the complex number  $2\sqrt{3} - 2i$  expressed in modulus–argument form?

**1**

- (A)  $4 \left( \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$  (B)  $4 \left( \cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$   
(C)  $4 \left( \cos(-\frac{\pi}{6}) + i \sin(-\frac{\pi}{6}) \right)$  (D)  $4 \left( \cos(-\frac{\pi}{3}) + i \sin(-\frac{\pi}{3}) \right)$

**QUESTION FOUR**

Which of the following is a correct primitive of  $\frac{1}{\sqrt{16 - 9x^2}}$  ?

**1**

- (A)  $\frac{1}{3} \sin^{-1} \left( \frac{3x}{4} \right)$  (B)  $\frac{1}{4} \sin^{-1}(3x)$   
(C)  $4 \sin^{-1} \left( \frac{3x}{4} \right)$  (D)  $\frac{4}{3} \sin^{-1} \left( \frac{4x}{3} \right)$

Exam continues next page ...

**QUESTION FIVE**

Given  $x^3 - 2xy^2 = 10$ , the correct result for  $\frac{dy}{dx}$  is:

1

(A)  $\frac{3x^2 - 2y^2}{4xy}$

(B)  $\frac{3x}{4}$

(C)  $\frac{3x}{4y}$

(D)  $\frac{2y^2 - 3x^2}{4xy}$

**QUESTION SIX**

Which of the following is the correct result for  $\int \frac{1}{x^2 - 4x + 8} dx$  ?

1

(A)  $2 \tan^{-1}(x - 2) + c$

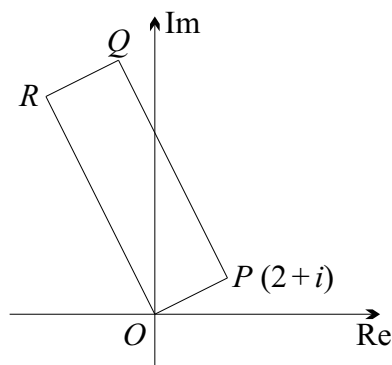
(B)  $\tan^{-1}\left(\frac{x - 2}{2}\right) + c$

(C)  $\frac{1}{2} \tan^{-1}(x - 2) + c$

(D)  $\frac{1}{2} \tan^{-1}\left(\frac{x - 2}{2}\right) + c$

**QUESTION SEVEN**

1



The diagram above shows a rectangle  $OPQR$  in the complex plane. The vertex  $P$  represents the complex number  $2 + i$  and the side  $OR$  is three times the length of  $OP$ . The complex number that corresponds to vertex  $R$  is:

(A)  $-6 - 3i$

(B)  $3 + 6i$

(C)  $6i - 3$

(D)  $3i - 6$

### QUESTION EIGHT

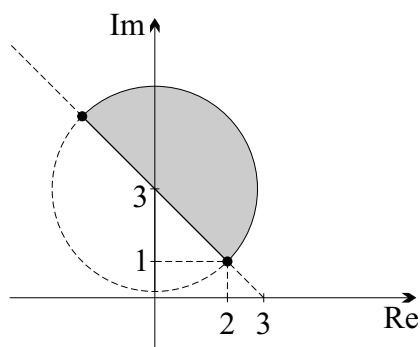
Which of the following statements about the inverse trigonometric functions is false?

1

- (A)  $\frac{d}{dx}(\sin^{-1} x + \cos^{-1} x) = 0$
- (B) The range of  $y = \cos^{-1} 3x$  is  $0 \leq y \leq \pi$ .
- (C) The inverse sine graph has a stationary point of inflection at the origin.
- (D)  $f(x) = \tan^{-1}(-x)$  is a purely decreasing function.

### QUESTION NINE

1



The shaded region in the diagram above illustrates the intersection of which of the following pairs of complex inequalities?

- (A)  $|z - 3| \leq 8$  and  $0 \leq \arg(z - 3i) \leq \frac{3\pi}{4}$
- (B)  $|z - 3i| \leq 8$  and  $0 \leq \arg(z - 3) \leq \frac{3\pi}{4}$
- (C)  $|z - 3| \leq 2\sqrt{2}$  and  $0 \leq \arg(z - 3i) \leq \frac{3\pi}{4}$
- (D)  $|z - 3i| \leq 2\sqrt{2}$  and  $0 \leq \arg(z - 3) \leq \frac{3\pi}{4}$

**QUESTION TEN**

A given function,  $f(x)$ , has period 2, that is  $f(x) = f(x + 2)$ .

**1**

Which of the following is certain to equal  $\int_{-1}^1 f(x) dx$  ?

(A)  $\int_0^2 f(x) dx$

(B) 0

(C)  $2 \int_0^1 f(x) dx$

(D)  $\int_1^3 f(x) dx$

————— End of Section I —————

Exam continues overleaf ...

## SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

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**QUESTION ELEVEN** (15 marks) Use a separate writing booklet.

**Marks**

(a) Let  $z = 3 - 4i$  and  $w = 5 + 2i$ .

(i) Find  $2z - 3w$ .

**1**

(ii) Find  $\overline{wz}$ .

**1**

(iii) Express  $\frac{w}{z}$  in the form  $a + ib$ , where  $a$  and  $b$  are real numbers.

**2**

(b) Given that  $z = 1 + 2i$  is a solution of the quadratic equation  $z^2 - kz + (6 + 2i) = 0$ , find the value of  $k$ .

**2**

(c) (i) Find the two square roots of  $8 + 6i$ .

**1**

(ii) Hence, or otherwise, solve the quadratic equation  $iz^2 + (1 + i)z + (-1 + 2i) = 0$ .

**2**

(d) A complex number  $z$  satisfies  $|z - 2 + 5i| = 1$ .

(i) Sketch the locus of  $z$  in the Argand diagram.

**1**

(ii) Hence find the minimum value of  $|z|$ .

**1**

(e) Let  $w_1 = 1 + i\sqrt{3}$  and  $w_2 = 2\sqrt{2}(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4})$ .

(i) Express  $w_1$  in modulus–argument form.

**1**

(ii) By considering the product  $w_1w_2$ , find the exact value of  $\sin \frac{7\pi}{12}$ .

**3**

**QUESTION TWELVE** (15 marks) Use a separate writing booklet.**Marks**

(a) Find  $\int_0^3 \frac{x-1}{\sqrt{x+1}} dx$  by using the substitution  $u = \sqrt{x+1}$ .

**3**

(b) (i) Find the values of  $A$ ,  $B$  and  $C$  such that

**2**

$$\frac{5x^2 + 7x + 8}{(2x-1)(x^2+4)} = \frac{A}{2x-1} + \frac{Bx+C}{x^2+4}.$$

(ii) Hence find  $\int \frac{5x^2 + 7x + 8}{(2x-1)(x^2+4)} dx$ .

**2**

(c) Use the  $t$ -formulae to find  $\int \frac{dx}{\sin x(1+\cos x)}$ .

**2**

(d) Use integration by parts to find  $\int e^{-x} \cos x dx$ .

**3**

(e) (i) Show that  $\frac{d}{dx} \ln(x + \sqrt{x^2+4}) = \frac{1}{\sqrt{x^2+4}}$ .

**1**

(ii) Hence, or otherwise, show that

**2**

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\sec^2 x}{\sqrt{\tan^2 x + 4}} dx = 2 \ln \left( \frac{\sqrt{5} + 1}{2} \right).$$

**QUESTION THIRTEEN** (15 marks) Use a separate writing booklet.**Marks**

(a) A locus in the Argand diagram is described by  $z\bar{z} = \left(\operatorname{Re}(z + 2)\right)^2$ . Determine the Cartesian equation of the locus. **2**

(b) (i) Differentiate  $f(x) = \tan^{-1} x + \tan^{-1} \frac{1}{x}$ , where  $x > 0$ . **1**

(ii) Hence find the exact value of  $\tan^{-1} 2 + \tan^{-1} \frac{1}{2}$ . **1**

(c) Use mathematical induction to prove that for all positive integer values of  $n$ : **3**  
 $(\cos \theta - i \sin \theta)^n = \cos(n\theta) - i \sin(n\theta)$

(d) Use the product to sum result  $\sin A \sin B = -\frac{1}{2}(\cos(A + B) - \cos(A - B))$  **2**  
to evaluate  $\int_0^{\frac{\pi}{4}} \sin 7x \sin 3x \, dx$ .

(e) Find the gradient of the tangent to the curve  $\frac{x^2}{9} + \frac{y^2}{4} = 1$  at the point  $\left(\frac{3\sqrt{2}}{2}, \sqrt{2}\right)$ . **2**

(f) Let  $u_n = \int_0^{\frac{\pi}{4}} \tan^{2n} \theta \, d\theta$ , where  $n \geq 0$ .

(i) Show that  $u_n = \frac{1}{2n-1} - u_{n-1}$ , where  $n \geq 1$ . **2**

(ii) Hence evaluate  $\int_0^{\frac{\pi}{4}} \tan^6 \theta \, d\theta$ . **2**

**QUESTION FOURTEEN** (15 marks) Use a separate writing booklet.**Marks**

- (a) (i) Use the results for
- $\sin(A \pm B)$
- to prove that

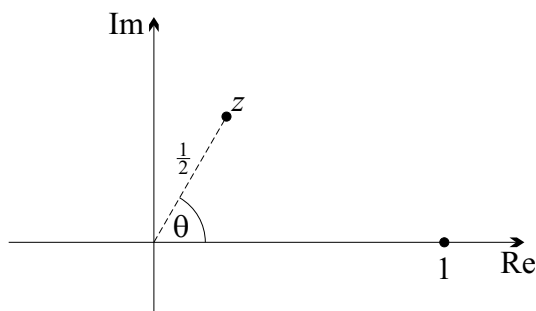
**1**

$$\sin C + \sin D = 2 \sin \frac{1}{2}(C + D) \cos \frac{1}{2}(C - D).$$

- (ii) Hence, or otherwise, find the general solution of
- $\sin 3x + \sin x = \cos x$
- .

**3**

- (b) Consider the sequence
- $1, z, z^2, z^3, \dots$
- , where
- $z = \frac{1}{2}(\cos \theta + i \sin \theta)$
- and
- $\theta$
- is acute.



- (i) Copy the Argand diagram given above into your answer booklet and add points to represent
- $z^2$
- and
- $z^3$
- .

**1**

- (ii) Explain why
- $\text{Im}(z^n) = (\frac{1}{2})^n \sin n\theta$
- .

**1**

- (iii) The series
- $1 + z + z^2 + z^3 + \dots$
- converges to a limit.

**3**

$$\text{Show that } \frac{1}{2} \sin \theta + \frac{1}{4} \sin 2\theta + \frac{1}{8} \sin 3\theta + \dots = \frac{2 \sin \theta}{5 - 4 \cos \theta}.$$

- (c) (i) Show that
- $\frac{1}{x^4 + 4} = \frac{x + 2}{8(x^2 + 2x + 2)} - \frac{x - 2}{8(x^2 - 2x + 2)}.$

**1**

- (ii) Consider the integral
- $I = \int_0^N \frac{1}{x^4 + 4} dx.$

**3**

$$\text{Show that } I = \frac{1}{16} \left( \log \left( \frac{N^2 + 2N + 2}{N^2 - 2N + 2} \right) + 2 \tan^{-1}(N + 1) + 2 \tan^{-1}(N - 1) \right).$$

- (iii) Hence evaluate
- $\lim_{N \rightarrow \infty} \int_{-N}^N \frac{1}{x^4 + 4} dx.$

**2**

\_\_\_\_\_ End of Section II \_\_\_\_\_

**END OF EXAMINATION**

B L A N K   P A G E

B L A N K   P A G E

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

NOTE :  $\ln x = \log_e x, \quad x > 0$



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FORM VI  
MATHEMATICS EXTENSION 2  
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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

CANDIDATE NUMBER: .....

**Question One**

A ☐ B ☐ C ☐ D ☐

**Question Two**

A ☐ B ☐ C ☐ D ☐

**Question Three**

A ☐ B ☐ C ☐ D ☐

**Question Four**

A ☐ B ☐ C ☐ D ☐

**Question Five**

A ☐ B ☐ C ☐ D ☐

**Question Six**

A ☐ B ☐ C ☐ D ☐

**Question Seven**

A ☐ B ☐ C ☐ D ☐

**Question Eight**

A ☐ B ☐ C ☐ D ☐

**Question Nine**

A ☐ B ☐ C ☐ D ☐

**Question Ten**

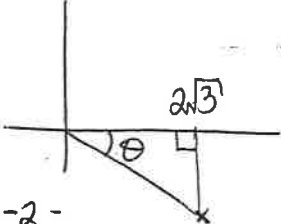
A ☐ B ☐ C ☐ D ☐

# 4U Half Yearly 2014

## Multiple Choice

①  $z = 2i$  or  $1+i$  (B) ✓

②  $\int e^{2x} \sin(e^{2x}) dx = -\frac{1}{2} \cos e^{2x}$  (D) ✓

③   $r^2 = (-2)^2 + (2\sqrt{3})^2 = 4 + 12 = 16$   
 $r = 4$   
 $\theta = -\tan^{-1}\left(\frac{2\sqrt{3}}{2}\right) = -\tan^{-1}(\sqrt{3}) = -\frac{\pi}{6}$  in  $4\cos(-\frac{\pi}{6})$  (C) ✓

④  $\int \frac{1}{\sqrt{16-9x^2}} dx = \frac{1}{3} \int \frac{1}{\sqrt{\left(\frac{4}{3}\right)^2 - x^2}} dx$   
 $= \frac{1}{3} \sin^{-1}\left(\frac{x}{\frac{4}{3}}\right) + C$   
 $= \frac{1}{3} \sin^{-1}\left(\frac{3x}{4}\right) + C$  (A) ✓

⑤  $x^3 - 2xy^2 = 10$   
 Differentiate implicitly  
 $3x^2 - 2y^2 - 2x(2y \frac{dy}{dx}) = 0$   
 $3x^2 - 2y^2 = 4xy \frac{dy}{dx}$   
 $\frac{dy}{dx} = \frac{3x^2 - 2y^2}{4xy}$  (A) ✓

⑥  $\int \frac{1}{x^2 - 4x + 8} dx = \int \frac{1}{(x-2)^2 + 4} dx$   
 $= \frac{1}{2} \tan^{-1}\left(\frac{x-2}{2}\right) + C$  (D) ✓

⑦  $P$  represents  $(2+i)$   
 rotation by  $90^\circ$  anticlockwise  $\Rightarrow$  multiplication by  $i$   
 enlarge by factor of 3 radially  $\Rightarrow \times 3$   
 $3i(2+i) = 6i - 3$  (C) ✓

⑧  $\frac{d}{dx}(\sin^{-1}x + \cos^{-1}x) = \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x^2}} = 0$  TRUE  
 $y = \cos^{-1} 3x$  stretch in  $x$  direction with factor  $\frac{1}{3}$   
 Range  $0 \leq y \leq \pi$  TRUE

$y = \sin^{-1} x$  Has point of inflection at origin but gradient is one is non stationary FALSE  
 $f(x) = \tan^{-1}(-x)$  Inverse tan reflected in  $y$ -axis  $\therefore$  decreasing TRUE (C) ✓

⑨  $|z - 3i| \leq 8$  radius  $= \sqrt{2^2 + 2^2} = \sqrt{8}$  (D) ✓  
 and  $0 \leq \arg(z-3) \leq \frac{3\pi}{4}$

⑩  $\int_1^1 f(x) dx = \int_1^3 f(x) dx$  since periodicity 2 (D) ✓

Total (10)

11 a)  $z = 3 - 4i$   $w = 5 + 2i$

(i)  $2z - 3w = 2(3 - 4i) - 3(5 + 2i)$   
 $= 6 - 8i - 15 - 6i$   
 $= -9 - 14i$  ✓

(ii)  $\overline{wz} = \overline{(3 - 4i)(5 + 2i)}$   
 $= \overline{15 - 20i + 6i - 8i^2}$   
 $= \overline{23 - 14i}$   
 $= 23 + 14i$  ✓

(iii)  $\frac{w}{z} = \frac{5 + 2i}{3 - 4i} \left( \times \frac{3 + 4i}{3 + 4i} \right)$  ✓  
 $= \frac{15 + 20i + 6i + 8i^2}{25}$   
 $= \frac{7 + 26i}{25} = \frac{7}{25} + \frac{26}{25}i$  (ie  $a = \frac{7}{25}$   $b = \frac{26}{25}$ )

(b)  $1 + 2i$  a solution  $z^2 - kz + (6 + 2i) = 0$   
 $\therefore$  satisfies LHS = 0  
 $(1 + 2i)^2 - k(1 + 2i) + (6 + 2i) = 0$  ✓  
 $-3 + 4i - k - 2ki + 6 + 2i = 0$   
 $(3 - k) + (6 - 2k)i = 0$   
 $\therefore k = 3$  ✓

(c) i) Let  $(a + bi)^2 = 8 + 6i$   
 $\therefore a^2 - b^2 = 8$  ①  
 $2ab = 6 \Rightarrow ab = 3$  ②

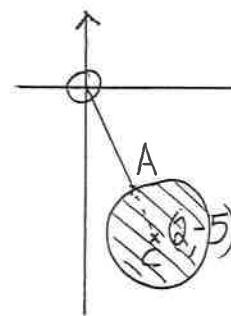
by inspection  $a = 3$   $b = 1$   
 or  $a = -3$   $b = -1$

$\therefore$  square roots are  $3 + i$  or  $-(3 + i)$  ✓

(ii)  $iz^2 + (1 + i)z + (-1 + 2i) = 0$   $\Delta = (1 + i)^2 - 4 \times i \times (-1 + 2i)$   
 $\therefore z = \frac{-(1 + i) \pm 3 + i}{2i}$  or  $\frac{-(1 + i) - (3 + i)}{2i}$   $= \frac{2i + 4i - 8i^2}{2i}$  ✓  
 $= \frac{2}{2i}$   $= \frac{-4 - 2i}{2i}$   
 $= \frac{1}{i}$   $= \frac{-2 - 0}{i} (x - i)$   
 $= -i$   $= \frac{-1 + 2i}{i} (x - i)$  ✓

d)

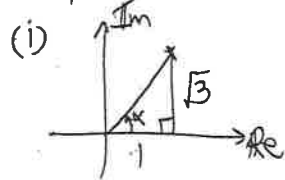
$|z - 2 + 5i| \leq 1$   $\odot$  Centre  $(2, -5)$  Radius 1  
 Inside Region



$OC = \sqrt{2^2 + 5^2}$   
 $= \sqrt{29}$

$\therefore \min |z| = \frac{OA}{\sqrt{29}} - 1$  ✓

e)  $\omega_1 = 1 + i\sqrt{3}$   $\omega_2 = 2\sqrt{2}\omega_0 \frac{\pi}{4}$



$$\omega_1 = r \cos x$$

$$r^2 = 1^2 + (\sqrt{3})^2$$

$$r = 2$$

$$\tan x = \sqrt{3}$$

$$x = \frac{\pi}{3}$$

$$\therefore \omega_1 = 2\omega_0 \frac{\pi}{3}$$

(ii)  $\omega_1 \omega_2 = 2\omega_0 \frac{\pi}{3} \times 2\sqrt{2}\omega_0 \frac{\pi}{4}$   
 $= 4\sqrt{2}\omega_0 (\frac{\pi}{3} + \frac{\pi}{4})$   
 $= 4\sqrt{2}\omega_0 (\frac{7\pi}{12})$  ✓

$$\omega_1 \omega_2 = (1 + i\sqrt{3}) \times (2 + 2i)$$
  
 $= 2 + 2\sqrt{3}i + 2i - 2\sqrt{3}$   
 $= 2 - 2\sqrt{3} + (2 + 2\sqrt{3})i$  ✓

$$\therefore 4\sqrt{2} \sin \frac{7\pi}{12} = 2 + 2\sqrt{3}$$
  
 $\therefore \sin \frac{7\pi}{12} = \frac{2 + 2\sqrt{3}}{4\sqrt{2}}$  ✓  
 $= \frac{1 + \sqrt{3}}{2\sqrt{2}}$   
 $= \frac{\sqrt{2} + \sqrt{6}}{4}$

15

(12) a)  $\int_0^3 \frac{x-1}{\sqrt{x+1}} dx$   
 $= \int_1^2 [(u^2-1)] \times 2 du$   
 $= \int_1^2 [2u^2 - 4] du$  ✓  
 $= \left[ \frac{2u^3}{3} - 4u \right]_1^2$   
 $= \left( \frac{2 \times 8}{3} - 8 \right) - \left( \frac{2}{3} - 4 \right)$   
 $= \frac{16}{3} - 4$   
 $= \frac{2}{3}$  ✓

$$u = \sqrt{x+1}$$
  
 $du = \frac{1}{2}(x+1)^{-\frac{1}{2}} dx$   
 $2du = \frac{dx}{\sqrt{x+1}}$  ✓

$$u^2 - 1 = x$$

|   |   |   |
|---|---|---|
| x | 0 | 3 |
| u | 1 | 2 |

b)  $\frac{5x^2 + 7x + 8}{(2x-1)(x^2+4)} = \frac{A}{2x-1} + \frac{Bx+C}{x^2+4}$   
 $= \frac{Ax^2 + 4A + (Bx+C)(2x-1)}{(2x-1)(x^2+4)}$   
 $= \frac{(A+2B)x^2 + (2C-B)x + (4A-C)}{(2x-1)(x^2+4)}$

Comparing coeffs

|       |              |   |
|-------|--------------|---|
| $x^2$ | $A + 2B = 5$ | ① |
| $x$   | $2C - B = 7$ | ② |
| $x^0$ | $4A - C = 8$ | ③ |

① + 2 × ②  $A + 4C = 19$  ④  
 ④ - ③  $17A = 51$   
 $A = 3$   
 $\therefore B = 1$   
 $\therefore C = 4$  ✓

$$\therefore \text{LHS} = \frac{3}{2x-1} + \frac{x+4}{x^2+4}$$

(ii)  $\int \frac{5x^2 + 7x + 8}{(2x-1)(x^2+4)} dx = \int \frac{3}{2x-1} + \frac{x}{x^2+4} + \frac{4}{x^2+4} dx$  ✓ One Constant Primitive  
 $= \frac{3}{2} \ln |2x-1| + \frac{1}{2} \ln(x^2+4) + 4 \times \frac{1}{2} \tan^{-1} \left( \frac{x}{2} \right) + C$   
 $= \frac{1}{2} \left( 3 \ln |2x-1| + \ln(x^2+4) + 4 \tan^{-1} \left( \frac{x}{2} \right) \right) + C$  ✓ These Constant

c)  $\int \frac{dx}{\sin x(1+\cos x)}$

Let  $t = \tan \frac{x}{2}$   
 $\therefore x = 2 \tan^{-1} t$   
 $dx = \frac{2dt}{1+t^2}$   
 $\sin x = \frac{2t}{1+t^2}$      $\cos x = \frac{1-t^2}{1+t^2}$

$$= \int \frac{\frac{2dt}{1+t^2}}{\frac{2t}{1+t^2} \left(1 + \frac{1-t^2}{1+t^2}\right)}$$

$$= \int \frac{dt}{t(1+\frac{1-t^2}{1+t^2})}$$

$$= \int \frac{1+t^2}{t(1+t^2+1-t^2)} dt$$

$$= \int \frac{1+t^2}{2t} dt$$

$$= \int \left(\frac{1}{2t} + \frac{t}{2}\right) dt$$

$$= \frac{1}{2} \ln|t| + \frac{t^2}{4} + C = \frac{1}{2} \ln|\tan \frac{x}{2}| + \frac{\tan^2(\frac{x}{2})}{4} + C \quad \checkmark$$

d)  $\int e^{-x} \cos x dx$

Let  $u = \cos x$      $\frac{du}{dx} = -\sin x$      $v = -e^{-x}$   
 $\frac{dv}{dx} = e^{-x}$

$$= (\cos x)(-e^{-x}) - \int (-e^{-x})(-\sin x) dx$$

$$= -e^{-x} \cos x - \int e^{-x} \sin x dx \quad \checkmark$$

Let  $u = \sin x$      $\frac{du}{dx} = \cos x$      $v = -e^{-x}$   
 $\frac{dv}{dx} = e^{-x}$

$$I = -e^{-x} \cos x - \left[ \sin x (-e^{-x}) - \int (-e^{-x})(\cos x) dx \right]$$

$$I = -e^{-x} \cos x + e^{-x} \sin x - \int e^{-x} \cos x dx \quad \checkmark$$

$$2I = \left[ e^{-x} (\sin x - \cos x) \right] + C_1$$

$$I = \frac{1}{2} e^{-x} (\sin x - \cos x) + C_2 \quad \checkmark$$

(e) (i)  $\frac{d}{dx} \ln(x + \sqrt{x^2+4}) = \frac{1}{x + \sqrt{x^2+4}} \times \left(1 + \frac{1}{2}(x^2+4)^{-\frac{1}{2}} \times 2x\right)$

$$= \frac{1}{x + \sqrt{x^2+4}} \times \left(1 + \frac{x}{\sqrt{x^2+4}}\right) \quad \checkmark \text{SHOW}$$

$$= \frac{1}{(x + \sqrt{x^2+4})} \times \left(\frac{\sqrt{x^2+4} + x}{\sqrt{x^2+4}}\right)$$

$$= \frac{1}{\sqrt{x^2+4}}$$

(ii)  $\int_{-\pi/4}^{\pi/4} \frac{\sec^2 x}{\sqrt{\tan^2 x + 4}} dx$

substitute  $u = \tan x$   
 $\therefore du = \sec^2 x dx$

|     |                 |                  |
|-----|-----------------|------------------|
| $x$ | $\frac{\pi}{4}$ | $-\frac{\pi}{4}$ |
| $u$ | 1               | -1               |

$$= \int_{-1}^1 \frac{du}{\sqrt{u^2+4}}$$

$$= \left[ \ln(u + \sqrt{u^2+4}) \right]_{-1}^1 \quad \checkmark$$

$$= \ln(1 + \sqrt{5}) - \ln(-1 + \sqrt{5})$$

$$= \ln\left(\frac{1+\sqrt{5}}{\sqrt{5}-1}\right)$$

$$= \ln\left[\frac{(1+\sqrt{5})(\sqrt{5}+1)}{5-1}\right] = \ln\left(\frac{(1+\sqrt{5})^2}{4}\right) \quad \checkmark \text{SHOW}$$

$$= \ln\left(\frac{1+\sqrt{5}}{2}\right)^2$$

$$= 2 \ln\left(\frac{1+\sqrt{5}}{2}\right)$$

$$(13) a) z\bar{z} = [\operatorname{Re}(z+2)]^2$$

$$\text{Let } z = x+iy. \text{ LHS} = z\bar{z} = (x+iy)(x-iy) = x^2+y^2$$

$$\text{RHS} = [\operatorname{Re}(x+2+iy)]^2 = (x+2)^2$$

$$\therefore x^2+y^2 = (x+2)^2$$

$$x^2+y^2 = x^2+4x+4$$

$$\therefore y^2 = 4(x+1)$$

$$b) i) f(x) = \tan^{-1} x + \tan^{-1}\left(\frac{1}{x}\right) \quad (x > 0)$$

$$f'(x) = \frac{1}{1+x^2} + \frac{1}{1+\left(\frac{1}{x}\right)^2} \times (-x^{-2})$$

$$= \frac{1}{1+x^2} + \frac{-x^2}{1+\frac{1}{x^2}} \times (x^2)$$

$$= \frac{1}{1+x^2} - \frac{1}{x^2+1}$$

$$= 0$$

$$(ii) \text{ Since } f'(x) = 0 \quad f(x) = \text{constant}$$

$$\text{Sub } x=1 \quad f(1) = \tan^{-1} 1 + \tan^{-1}\left(\frac{1}{1}\right)$$

$$= \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}$$

$$\text{hence } f(x) = \frac{\pi}{2}$$

$$\text{hence } \tan^{-1} 2 + \tan^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{2}$$

$$c) \text{ Prove } (\cos \theta - i \sin \theta)^n = \cos n\theta - i \sin n\theta$$

STEP A

Show for  $n=1$

$$\text{LHS} = (\cos \theta - i \sin \theta)^1$$

$$\text{RHS} = \cos \theta - i \sin \theta$$

$$\therefore \text{LHS} = \text{RHS} \quad \text{Result is true for } n=1.$$

STEP B

Assume result is true for  $n=k$ ,  $k$  positive integer

$$\text{ie } (\cos \theta - i \sin \theta)^k = \cos k\theta - i \sin k\theta$$

Attempt to show for  $n=k+1$

$$\text{ie } (\cos \theta - i \sin \theta)^{k+1} = \cos (k+1)\theta - i \sin (k+1)\theta$$

$$\text{LHS} = (\cos \theta - i \sin \theta)^k \times (\cos \theta - i \sin \theta)$$

$$= (\cos k\theta - i \sin k\theta) \times (\cos \theta - i \sin \theta) \quad \text{using assumption}$$

$$= \cos k\theta \cos \theta + i \sin k\theta \sin \theta - i \sin k\theta \cos \theta - i \cos k\theta \sin \theta$$

$$= (\cos k\theta \cos \theta - \sin k\theta \sin \theta) - i (\sin k\theta \cos \theta + \cos k\theta \sin \theta)$$

$$= \cos (k+1)\theta - i \sin (k+1)\theta \quad \text{using compound angle formulae}$$

$$= \text{RHS}$$

$\therefore$  If result true for  $n=k$ , also true for  $n=k+1$ .

STEP C

By principle of mathematical induction, result proven for all positive integer  $n$ .

$$d) \text{ Given } -2 \sin A \sin B = \cos (A+B) - \cos (A-B)$$

$$\int_0^{\pi/4} \sin 7x \sin 3x \, dx$$

$$= \int_0^{\pi/4} -\frac{1}{2} [\cos 10x - \cos 4x] \, dx$$

$$= -\frac{1}{2} \left[ \frac{1}{10} \sin 10x - \frac{1}{4} \sin 4x \right]_0^{\pi/4}$$

$$= -\frac{1}{2} \left[ \left( \frac{1}{10} \sin \frac{5\pi}{2} - \frac{1}{4} \sin \pi \right) - (0) \right]$$

$$= -\frac{1}{2} \times 0$$

$$e) \frac{x^2}{9} + \frac{y^2}{4} = 1 \quad \text{Diff implicitly wrt } x$$

$$\frac{2x}{9} + \frac{2y}{4} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-2x/9}{2y/4} = -\frac{4x}{9y} \quad \checkmark$$

$$\text{Gradient @ } \left(\frac{3\sqrt{2}}{2}, \sqrt{2}\right) \quad m = \frac{-4\left(\frac{3\sqrt{2}}{2}\right)}{9\sqrt{2}}$$

$$= -\frac{2}{3} \quad \checkmark$$

$$f(i) u_n = \int_0^{\pi/4} \tan^{\frac{2n}{2}} \theta \, d\theta$$

$$= \int_0^{\pi/4} \tan^{\frac{2n-2}{2}} \theta (\sec^2 \theta - 1) \, d\theta$$

$$u_n = \int_0^{\pi/4} \tan^{\frac{2n-2}{2}} \theta \sec^2 \theta \, d\theta - \int_0^{\pi/4} \tan^{\frac{2n-2}{2}} \theta \, d\theta \quad \checkmark$$

$$u_n = \left[ \frac{\tan^{\frac{2n-1}{2}} \theta}{\frac{2n-1}{2}} \right]_0^{\pi/4} - u_{n-1}$$

$$= \left[ \frac{\tan^{\frac{2n-1}{2}} \left(\frac{\pi}{4}\right) - \tan^{\frac{2n-1}{2}} 0}{\frac{2n-1}{2}} \right] - u_{n-1} \quad \checkmark$$

$$u_n = \frac{1}{2n-1} - u_{n-1} \quad (n \geq 1)$$

$$(ii) \int_0^{\pi/4} \tan^6 \theta \, d\theta = \frac{1}{2 \times 3 - 1} - \int_0^{\pi/4} \tan^4 \theta \, d\theta$$

$$= \frac{1}{5} - \left[ \frac{1}{2 \times 2 - 1} - \int_0^{\pi/4} \tan^2 \theta \, d\theta \right] \quad \checkmark$$

$$= \frac{1}{5} - \frac{1}{3} + \left[ \frac{1}{2 \times 1 - 1} - \int_0^{\pi/4} d\theta \right]$$

$$= \frac{1}{5} - \frac{1}{3} + 1 - \left[ \theta \right]_0^{\pi/4}$$

$$= \frac{1}{5} - \frac{1}{3} + 1 - \frac{\pi}{4} = \frac{13}{15} - \frac{\pi}{4} \quad \checkmark$$

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$$④ a) (i) \sin(A+B) = \sin A \cos B + \cos A \sin B \quad ①$$

$$\sin(A-B) = \sin A \cos B - \cos A \sin B \quad ②$$

$$① + ② \quad \sin(A+B) + \sin(A-B) = 2 \sin A \cos B \quad *$$

$$\text{Let } C = A+B \quad ③$$

$$D = A-B \quad ④$$

$$③ + ④: C+D = 2A$$

$$A = \frac{1}{2}(C+D)$$

$\therefore$  substituting into \*

$$\sin C + \sin D = 2 \sin \frac{1}{2}(C+D) \cos \frac{1}{2}(C-D)$$

$$③ - ④: C-D = 2B$$

$$B = \frac{1}{2}(C-D)$$

✓ SHOW

$$(ii) \sin 3x + \sin x = \cos x$$

$$\therefore \text{Let } C = 3x \quad D = x \Rightarrow \frac{1}{2}(C+D) = 2x, \quad \frac{1}{2}(C-D) = x$$

$$\therefore 2 \sin 2x \cos x = \cos x \quad \checkmark$$

$$2 \sin 2x \cos x - \cos x = 0$$

$$\cos x (2 \sin 2x - 1) = 0$$

$$\cos x = 0 \quad \text{or} \quad \sin 2x = \frac{1}{2}$$

$$x = \frac{(2n+1)\pi}{2} \quad \text{where } n \in \mathbb{Z} \quad \checkmark$$

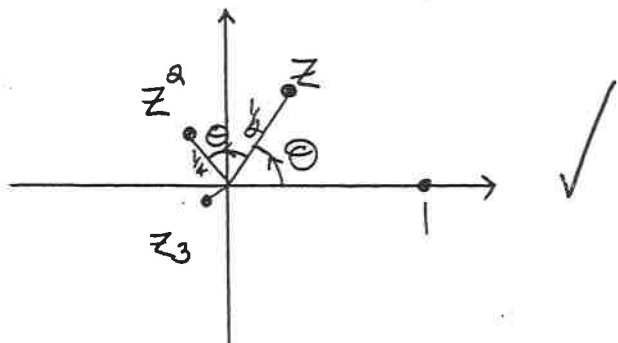
$$\text{OR} \quad 2x = 2n\pi + \frac{\pi}{6} \quad \text{or} \quad (2n+1)\pi - \frac{\pi}{6}$$

$$[\text{Alternative: } 2x = n\pi + (-1)^n \frac{\pi}{6}]$$

$$x = n\pi + \frac{\pi}{12} \quad \text{or} \quad n\pi + \frac{5\pi}{12} \quad \checkmark$$

$$[\text{Alternative: } x = \frac{n\pi}{2} + (-1)^n \frac{\pi}{12}]$$

b)(i)



(ii)  $z = \frac{1}{2} \cos \theta + i \sin \theta$   $\therefore$  multiplication by  $z$  halves the modulus and adds  $\theta$  to the argument ✓

$$z^n = \left(\frac{1}{2}\right)^n [\cos(\theta + \theta + \dots + \theta) + i \sin(\theta + \theta + \dots + \theta)]$$

$$= \left(\frac{1}{2}\right)^n (\cos n\theta + i \sin n\theta) \quad \therefore \operatorname{Im}\{z^n\} = \left(\frac{1}{2}\right)^n \sin n\theta$$

(iii)  $1 + z + z^2 + z^3 + \dots$  infinite GP  $a=1$   $r=z$

$$S_\infty = \frac{a}{1-r}$$

$$= \frac{1}{1-z}$$

$$= \frac{1}{1 - \frac{1}{2} \cos \theta}$$

$$= \frac{2}{2 - \cos \theta}$$

$$= \frac{2}{(2 - \cos \theta) - i \sin \theta} \times \frac{(2 - \cos \theta) + i \sin \theta}{(2 - \cos \theta) + i \sin \theta}$$

$$= \frac{(4 - 2\cos \theta) + i 2\sin \theta}{(4 - 4\cos \theta + \cos^2 \theta) + \sin^2 \theta}$$

$$= \frac{(4 - 2\cos \theta) + i 2\sin \theta}{5 - 4\cos \theta} \quad (*)$$

$$\therefore \operatorname{Im}\{S_\infty\} = \frac{2\sin \theta}{5 - 4\cos \theta}$$

from (i)  $\operatorname{Im}\{1 + z + z^2 + z^3 + \dots\} = 0 + \frac{1}{2}\sin \theta + \frac{1}{4}\sin 2\theta + \frac{1}{8}\sin 3\theta + \dots$

hence  $\frac{1}{2}\sin \theta + \frac{1}{4}\sin 2\theta + \frac{1}{8}\sin 3\theta + \dots = \frac{2\sin \theta}{5 - 4\cos \theta}$

c)(i)  $\frac{x+2}{8(x^2+2x+2)} - \frac{x-2}{8(x^2-2x+2)}$

$$= \frac{(x+2)(x^2-2x+2) - (x-2)(x^2+2x+2)}{8(x^2+2x+2)(x^2-2x+2)}$$

$$= \frac{(x^3-2x^2+4x+4) - (x^3+2x^2-4x-4)}{8(x^4-2x^3+2x^3-4x^2+4x+2x^2-4x+4)}$$

$$= \frac{8}{8(x^4+4)} \quad \checkmark$$

(ii)  $\int_0^N \frac{1}{(x^4+4)} dx = \int_0^N \frac{x+2}{8(x^2+2x+2)} - \frac{x-2}{8(x^2-2x+2)} dx$

$$= \frac{1}{8} \int_0^N \frac{(x+1)+1}{(x+1)^2+1} - \frac{(x-1)-1}{(x-1)^2+1} dx \quad \checkmark$$

$$= \frac{1}{8} \int_0^N \frac{x+1}{(x+1)^2+1} + \frac{1}{(x+1)^2+1} - \frac{(x-1)}{(x-1)^2+1} + \frac{1}{(x-1)^2+1} dx$$

$$= \frac{1}{8} \left[ \frac{1}{2} \ln(x+1)^2 + \tan^{-1}(x+1) - \frac{1}{2} \ln(x-1)^2 + \tan^{-1}(x-1) \right]_0^N$$

$$= \frac{1}{8} \left[ \frac{1}{2} \ln \left( \frac{x^2+2x+2}{x^2-2x+2} \right) + \tan^{-1}(x+1) + \tan^{-1}(x-1) \right]_0^N$$

$$= \frac{1}{16} \left[ \ln \left( \frac{N^2+2N+2}{N^2-2N+2} \right) + 2\tan^{-1}(N+1) + 2\tan^{-1}(N-1) - \ln \left( \frac{2}{2} \right) + 2\tan^{-1}1 + 2\tan^{-1}(-1) \right]$$

$$= \frac{1}{16} \left[ \ln \left( \frac{N^2+2N+2}{N^2-2N+2} \right) - 2\tan^{-1}(N+1) + 2\tan^{-1}(N-1) \right] \quad \checkmark$$

but  $\tan^{-1}x$  an odd function  
 $\therefore \tan^{-1}1 + \tan^{-1}(-1) = 0$   
 or  $\frac{\pi}{4} + (-\frac{\pi}{4}) = 0$

(iii)  $f(x) = \frac{1}{x^4+4}$  is an even function, hence  $\int_{-N}^N \frac{1}{x^4+4} dx = 2 \int_0^N \frac{1}{x^4+4} dx$

$$\lim_{N \rightarrow \infty} \int_{-N}^N \frac{1}{x^4+4} dx = 2 \times \lim_{N \rightarrow \infty} \int_0^N \frac{1}{x^4+4} dx \quad \checkmark$$

$$= 2 \times \frac{1}{16} \left[ \log 1 + 2\left(\frac{\pi}{2}\right) + 2\left(\frac{\pi}{2}\right) \right]$$

$$= \frac{1}{8} \times 2\pi$$

$$= \frac{\pi}{4} \quad \checkmark$$

since  $\log \left( \frac{N^2+2N+2}{N^2-2N+2} \right) = \log \left( \frac{1+\frac{2}{N}+\frac{2}{N^2}}{1-\frac{2}{N}+\frac{2}{N^2}} \right)$  as  $N \rightarrow \infty$   
 $\frac{2}{N}, \frac{2}{N^2} \rightarrow 0$   
 $\tan^{-1}(N+1)$  and  $\tan^{-1}(N-1) \rightarrow \frac{\pi}{2}$

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